Seyed Mohammad Amin Rajaei

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EDUCATION

Isfahan University of Technology (IUT) — **Isfahan, Iran**
Bachelor of Science in Computer Engineering — **2023 – Present**
Expected Graduation: Summer 2027
GPA: 16.59/20 (3.43/4.0 estimated via WES iGPA calculator)
Relevant Coursework:
Fundamentals of Machine Learning (in progress), Computational Intelligence (Deep Learning), Computational Intelligence (Deep Learning) Lab (in progress), Artificial Intelligence, Algorithm Design, Data Structures, Linear Algebra (in progress), Probability & Statistics

RESEARCH INTERESTS

Computer Vision, Deep Learning, Machine Learning

HONORS AND AWARDS

Ranked in the **top 0.7% nationwide** among all 147,317 Engineering and Mathematics applicants in Iran's National University Entrance Exam (Konkur), 2023

PROJECTS

Each project includes a research-style technical write-up in its GitHub README (theory, methodology and experimental analysis).

Vision Transformer Model from Scratch for Retinal OCT Disease Classification
Developed a Vision Transformer (ViT) model from scratch for retinal OCT image classification, implementing patch embeddings, multi-head self-attention, Transformer encoder blocks, and an MLP classification head.

CNN vs. Fully Connected Networks for Handwritten Digit Classification on MNIST
Explored fundamental deep learning concepts for computer vision by comparing a convolutional neural network (CNN) with a fully connected neural network for handwritten digit classification on the MNIST dataset. Achieved 99.63% test accuracy with the CNN, compared with 98.23% with the fully connected baseline.

Titanic Survival Prediction: Logistic Regression vs. Tree-Based Models
A Machine Learning Project, Including Data visualization, Data Preprocessing, Feature Engineering, Pipeline, Hyperparameter Optimization (improving the performance of tree-based models by 4%), Model Evaluation and Analysis. Comparing 3 different models. Test accuracy :
Logistic Regression: 83.2% Decision Tree: 82.1% Random Forest: 81.5%

SMS Spam Detection with NLP: Comparing Classical ML Classifiers
Developed a machine learning pipeline for SMS spam classification using natural language processing (NLP), comparing three classical classifiers. Test accuracy:
Logistic Regression: 97.3% Decision Tree: 94.1% Random Forest: 90.9%

Other Projects:

• **Plants vs. Zombies online multiplayer game in C++ using Qt Framework, Final project of advanced programming course at IUT**

- **RISC-V architecture simulator and assembler: Final project of Computer Architecture & Organization course at IUT**
- **Space War: cmd-based local multiplayer combat Game in C, Final project of basic programming course at IUT**

EXPERIENCE

AI Automation Intern **Summer 2026**

Institute of Artificial Intelligence, Isfahan University of Technology
Under the supervision of Dr. Hamidreza Hakimdavoodi

- Built an n8n workflow that uses an LLM API provided by Sharif University of Technology to score presentations and papers for the Iranian Orthopedic AI Congress across scientific quality, clinical impact, slides, speaker performance, timing, and Q&A, producing justified scores and rankings to support professors' evaluations.
- Built Telegram and other social media chatbots in n8n, integrated with platform APIs to automate message handling. one bot answers strictly from a CSV knowledge base, with response constraints that keep answers within the provided information.

TECHNICAL SKILLS

Programming: Python, C++
ML & DL Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib, seaborn
Computer Vision: Image Classification, Object Detection (YOLO, Faster R-CNN), Vision Transformers (ViT), Semantic & Instance Segmentation, Image Preprocessing, Data Augmentation, Model Evaluation (mAP, IoU, Dice Score, mIoU)
Deep Learning: Neural Networks, CNNs, Transfer Learning, Fine-tuning, Feature Extraction, Model Regularization (Dropout, Weight Decay, Batch Normalization), Learning Rate Scheduling and Optimization
Machine Learning: Data preprocessing, Feature engineering, Model evaluation
Tools: Git & GitHub, n8n, Jupyter Notebook, Linux

LANGUAGES

English: Upper-Intermediate (IELTS Preparation)
Persian: Native